

Ideation Phase

Define the Problem Statements

Date	26 June 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - who are they?	Describe the customer and their attributes here
I'm trying to	List their outcome or "job" the care about - what are they trying to achieve?	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way - what bothers them most?	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists - what needs to be solved?	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view - how does it impact them emotionally?	Describe the emotions the result from experiencing the problems or barriers

Reference: <https://miro.com/templates/customer-problem-statement/>

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a small-to-medium scale	decide which crop to grow each season	I can't reliably tell which	I lack access to an easy, data-driven	uncertain, financially exposed, and reliant on guesswork instead of

	farmer	for the best possible yield	crop suits my soil's N, P, K, pH levels and the season's temperature, humidity, and rainfall	tool that analyzes my soil and climate readings and recommends the most suitable crop	evidence
PS-2	a farmer or agronomist who already has a crop in mind	confirm whether my current soil and climate conditions are actually suitable for that crop	I have no quick way to check my soil and climate parameters against the crop's ideal requirements before investing time and money	there's no accessible system that evaluates my inputs against known crop-environment compatibility data	anxious about wasting resources on a crop that may underperform or fail
PS-3	an agricultural researcher or policymaker	analyze patterns in crop-environment relationships across regions to plan sustainable farming strategies	I lack a consolidated platform that aggregates soil, climate, and crop production data for analysis	existing agricultural data is scattered, manually maintained, and not structured for cross-crop or cross-region comparison	limited in my ability to make timely, evidence-based recommendations for sustainable agriculture